

MACHINE EARNING

Autonomous paper trading agent for the Alpaca hackathon



A Codex-based trading workflow that screens stocks, researches ideas, and makes paper trades.

SYSTEM

Machine Earning

MODE

Paper Trading

AGENT

Codex

DATA

Alpaca API

OBJECTIVE

Find edges. Test ideas.
Manage risk. Keep learning.

STAGE 0 AND STAGE 1



STAGE 0 — MARKET DATA

- Gets US stock universe from Alpaca
- Loads snapshots and daily bars
- Validates data and stores it locally



STAGE 1 — CANDIDATE DISCOVERY

- Applies statistical signals
- Adds community and expert attention
- Builds a shortlist for deeper research

LANES (EXAMPLES)

VOLUME ANOMALY

MOMENTUM BREAKOUT

STRETCHED REVERSAL

VOLATILITY EXPANSION

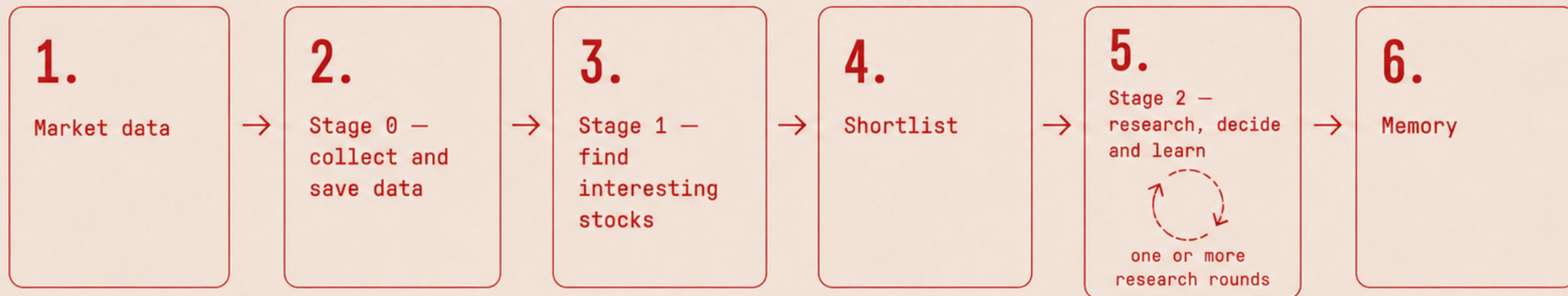
COMMUNITY & EXPERT

... AND WILDCARD

TYPICAL FUNNEL (EXAMPLE)



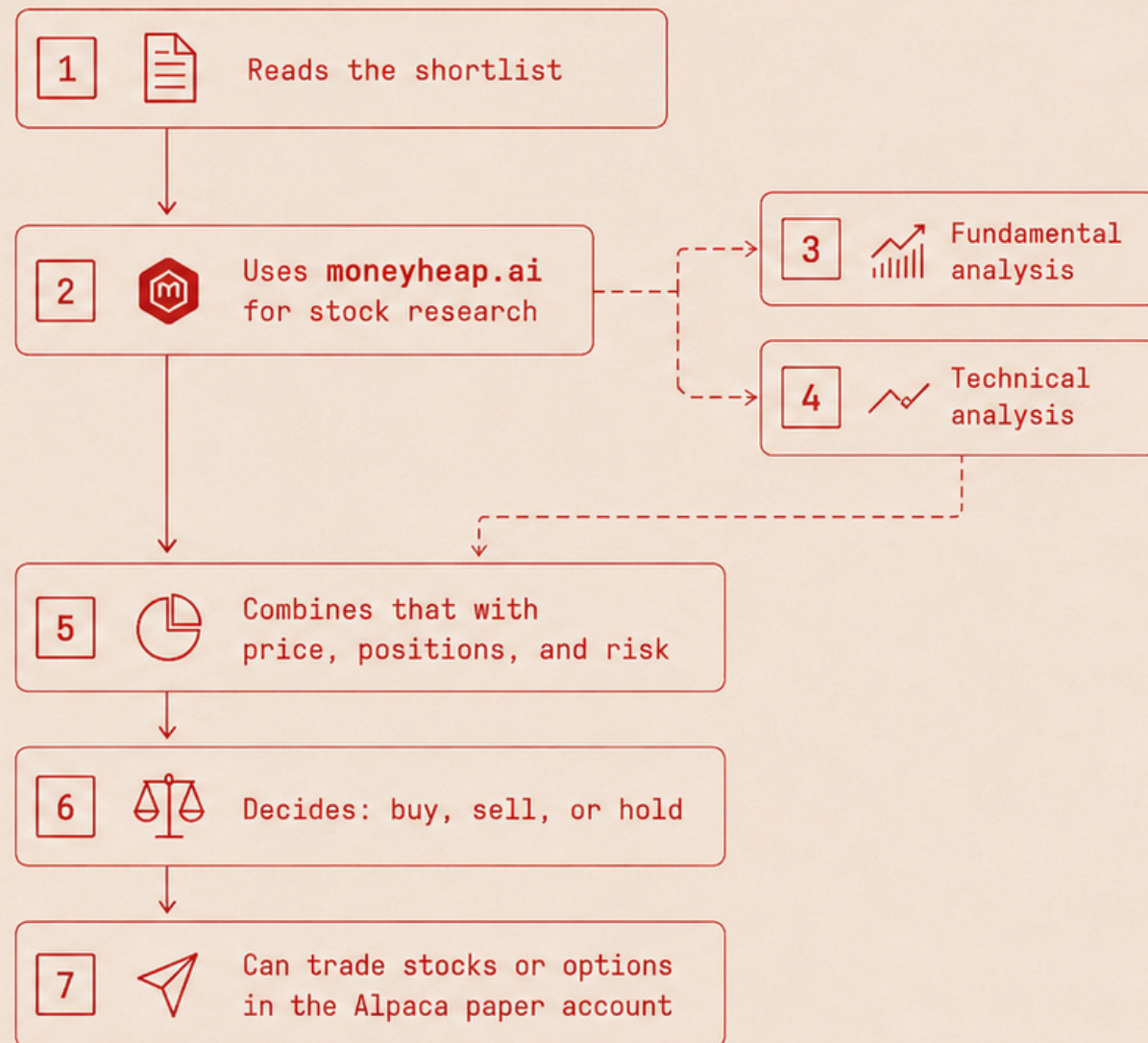
HOW IT IS STRUCTURED



Scripts do the data work. The Codex skill does the trading reasoning.

STAGE 2

In-depth research rounds
on shortlisted stocks



GHOST TRADES



REAL TRADE

- ✓ The agent makes a real paper trade.



GHOST ALTERNATIVES

- ✓ It also records a few realistic alternatives.
- ✓ These alternatives are simulated only.



LESSONS LEARNED

- ✓ Later it compares the real trade with the ghost trades.
- ✓ Then it writes small lessons for future runs.



This helps the system learn not only from mistakes, but also from missed better choices.

MEMORY CONTRACT

What the agent remembers between runs.



1. Previous trades

What the agent bought, sold, closed, or kept.

Gives context from earlier runs.



2. Ghost trades

Alternative trades that were simulated only.

Used to compare real vs. unrealized choices.



3. Logs

Daily summaries, actions, reasoning, and outcomes.

A trace of what happened each run.



4. Lessons learned

Short durable lessons updated over time.

Used as guidance in later decisions.

remember —————> compare —————> learn

The agent reads this before acting, and updates it after each cycle.

DASHBOARD

Agent-generated view of the paper account and daily activity.



1. Today's summary

Daily note written by the agent.



2. Portfolio and cash

Equity, cash, positions, and exposure.



3. Broker sync

Market session, buying power, and fills.



4. Activity tabs

Research, ghost trades, activity, and portfolio pages.

Machine Earning
paper trader

WORKSPACE

- Overview
- Portfolio
- Research
- Ghost trades
- Activity
- About

Workspace / Overview

Last synced 03 Sep 2026, 21:10 CESTRefresh

STAGE 2 - PAPER ACCOUNT

Markets are open; I'm monitoring the book.

Live paper-account state from Alpaca, matched to today's agent log.

SESSION - 2026-09-03

Agent note

memory/logs/2026-09-03-summary.md

Today's summary

Today was busy but fairly controlled. I didn't find anything convincing enough to buy, so I mostly stayed put. Sold HOOD 25/75 shares at \$121.984. Closed MSTR one of two Sep. 11 \$125/\$135 call spreads for a \$6.90 credit. I kept the remaining GAP, GTLB, HOOD, MMED, MU, RBLX and SLB positions in place. The account is currently at \$102,328.97, with the overall return at +\$2,328.97. HOOD helped most at +\$937.60, while GTLB was the main drag at -\$173.25.

Open daily summary

- 2026-09-03 21:09:24 Europe/Amsterdam

EQUITY

\$102,328.97

Since start - +\$2,328.97 (2.329%)

CASH AVAILABLE

\$53,950.09

Buying power - \$348,948.44

LONG MARKET VALUE

\$49,439.88

11 broker rows reconciled

OPEN POSITIONS

7 stocks + 2 option spreads

11 broker rows - 0

BROKER SYNC

Trading session

03 Sep 2026, 15:10 EDT - Live Alpaca paper account - ACTIVE

NEXT CLOSE

03 Sep 2026, 16:00 EDT

NEXT OPEN

04 Sep 2026, 09:30 EDT

BUYING POWER

\$348,948.44

OPTIONS BUYING POWER

\$75,401.74

Market open

TODAY'S FILLS

4 EXECUTIONS - 2 ORDERS

Live paper account

Synced 03 Sep 2026, 21:10 CEST

LIVE DASHBOARD | machine-earning.adelzakirov.chatgpt.site

Built automatically from agent logs and Alpaca paper-account state.

MACHINES LEARN. PORTFOLIOS EARN.



WHAT WE BUILT

An autonomous investing agent that:

- Scans the entire US market
- Finds high-probability opportunities
- Runs in-depth research
- Decides and trades
- Monitors, learns and improves



WHY IT MATTERS

Markets never sleep.

Our agent works 24/7 to:

- Remove emotion
- Stay disciplined
- Use data, not noise
- Keep getting better with experience



WHAT'S NEXT

- More research rounds
- Smarter risk management
- Richer data and signals
- Live trading when ready



The agent already keeps memory, learns from the past, and tracks what could've been.
It's built to compound.



Paper account today.
Real edge tomorrow.
Let the machines earn.



Built for the
Alpaca Hackathon.
Thanks for watching.